Brandon Ling-xuan (Wade-Giles: hsuan) Chen
Researcher, Independent
Student, Island School, Hong Kong
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Taiwan and Hong Kong

Education

International Baccalaureate: Diploma Program – Island School, approx. June 2028

- Higher Level: Mathematics Analysis and Approaches, Physics, Economics
- Standard Level: Chemistry, Chinese B, English Language and Literature

Publications

- Shadow-Edge Timing for Urban Positioning with Low-Earth-Orbit Satellites - **Working Paper**
  Brandon Ling-xuan Chen

- Lightweight-Learning Assisted Pseudorange Reweighting for Smartphone GNSS in Urban Canyons - **Conference Peer Review**
  Brandon Ling-xuan Chen, Derek Hoi Wah Ng, Li-Ta Hsu

- Altitude-Dependent Geolocation Effects on Typhoon-Relative FORMOSAT-7/COSMIC-2 Radio Occultation Sampling and Refractivity Compositing - **Journal Revision Process**
  Brandon Ling-xuan Chen

- Altitude-Dependent Geolocation Effects on Typhoon-Relative FORMOSAT-7/COSMIC-2 Radio Occultation Sampling - **Accepted, TASTI 2026**
  Brandon Ling-xuan Chen

- Uncertainty-Aware 3D-Mapping-Aided GNSS with Probabilistic Building Models- **Journal Peer Review**
  Brandon Ling-xuan Chen

- The Value of Information in Smartphone 3D-Mapping-Aided GNSS - **Journal Peer Review**
  Brandon Ling-xuan Chen

Experience and Appointments

- Sept. 2026-Present: **Research Assistant**, Department of Aeronautical and Aviation Engineering, Hong Kong Polytechnic University
  - Research Assistant at the Intelligent Positioning and Navigation Laboratory (IPNL). Mentored, supervised, recruited and appointed by Professor Li-Ta Hsu
  - Supervised by Professor Li-Ta Hsu
  - Promoted from Independent Researcher by Prof. Li-Ta Hsu, Independent Researcher → Research Assistant
  - Work focuses on: Positioning, Navigation and Timing (PNT); Urban GNSS; AI/ML GNSS; 3D-Mapping-Aided GNSS; Smartphone and Wearables GNSS; Low-Earth Orbit (LEO) Positioning

- Apr 2026-Sept. 2026: Independent Researcher, Department of Aeronautical and Aviation Engineering, Hong Kong Polytechnic University

- Apr. 2026-Sept. 2026: Hardware and Software Engineer, Industrial Centre, Hong Kong

Polytechnic University

Mechanical and Electronic Engineer working in CAD under the Stairio R&D Project. Appointed and supervised by Dr Max Lee Jwo-lem and Professor Li-Ta Hsu

Department of Mechanical Engineering and Department of Aeronautical and Aviation Engineering

- 2025-2026: Team Lead, Island School Typhoon Engineers, MIT EWB
  Team Lead within MIT'S EWB Competition for Science and Engineering
  Second Place Team in HK, Second in HK as an individual (90.32%)

- 2024-Present: Tutor in STEM, Self-Employment
  Tutored 15+ students in science and mathematics

- 2023-Present: Assistant Coach, Ortega Tennis Ltd.
  3+ years of Tennis Coaching
- 2023-Present: Musician, Self-Employment
  Freelance Musician with experience in both bands and as individual

Honors

Second Place in Hong Kong, Individual – 90.32%, ***Massachusetts Institute of Technology***

Second Place in Hong Kong, Team, ***Massachusetts Institute of Technology***

Certificate of Science Commendation, ***Island School***

Spirit Award for Dedication to Extracurricular Activities, ***Island School***

Professional and Academic Service

Member of the Institute of Navigation (2026-Present)
Student Assistant at the International Conference on Aerospace System Science Engineering (ICASSE) 2026
Editor-in-Chief, Island School Science Journal
STEM Committee Member, Island School
Head of Charity, Island School
Founder and President of Physics and Astrophysics Society, Island School
Teaching Assistant – Space Science, Island School Physics Department
Syllabus Support – Mathematics MYP + IGCSE, Island School Mathematics Department